

Disclaimer: This is an independent educational case study created as part of my “100 Days Quality Management Challenge.” SpaceX is a separate organisation. This is not an official SpaceX project, report, collaboration, or representation. The analysis uses publicly available information and general Quality Engineering principles. No confidential or proprietary SpaceX information is used.

THE QUALITY CHALLENGE

In reusable launch systems, a failure or unexpected deviation during testing is not simply a pass/fail event. It can provide important engineering evidence about design, hardware, software, process and reliability.

The Quality Engineering objective is to identify the failure mechanism, correct the underlying cause, verify the corrective action and prevent recurrence.

ILLUSTRATIVE QUALITY SCENARIO — NOT AN ACTUAL SPACEX INCIDENT

During repeated testing of a reusable launch-system component, performance gradually changes across test cycles and an unexpected failure or deviation occurs.

Core Investigative Questions:

What happened? • Why did it happen? • What is the risk? • What needs to change? • Did the corrective action work?

PROPOSED QUALITY SOLUTION

- Detect & Document:** Capture the failure, test conditions and objective evidence.
- Verify the Data:** Confirm the measurement and test setup are reliable.
- Analyse the Failure:** Use FMEA, FTA, reliability data and root-cause analysis as appropriate.
- Correct the Cause:** Improve the design, process, software/hardware interaction or test method.
- Re-test & Verify:** Repeat relevant verification and qualification activities.
- Prevent Recurrence:** Update risk assessments, configuration records and lessons learned.

QUALITY ENGINEERING APPROACH

INVESTIGATION & RESOLUTION WORKFLOW

Detect • Collect Evidence • Analyse • Identify Root Cause • Correct • Re-test • Verify • Prevent Recurrence

Core QE Tools & Contributions:

FMEA	Systematically evaluates potential failure modes and severity.
FTA	Deductive fault tree analysis to trace failure pathways.
Root Cause Analysis	Uncovers fundamental underlying systemic triggers.
Reliability Analysis	Assesses lifecycle degradation and failure probabilities.
Verification & Validation	Confirms requirements and corrective efficacy.
Corrective Action	Implements permanent engineering fixes.
Configuration Control	Tracks hardware and software baseline modifications.
Lessons Learned	Institutionalizes knowledge across programs.

WHY THIS APPROACH MATTERS FOR REUSABLE LAUNCH SYSTEMS

Publicly available SpaceX information describes an iterative development approach involving testing, learning from test outcomes and making engineering improvements.

Iterative Feedback Loop:

Test • Failure/Deviation • Data • Root Cause • Design/Process Improvement • Re-test • Reliability Improvement

Note: This does not claim an internal SpaceX QA procedure, nor does it use proprietary statistics.

EXPECTED QUALITY IMPACT

- Earlier identification of failure risks
- Better understanding of failure mechanisms
- Verified corrective actions
- Reduced recurrence risk and improved reliability

“A failure during testing is not just a setback — when analysed correctly, it becomes engineering data for building a more reliable system.”